SECURITY



# Confidential Storage

## Extending the Confidential-Computing Trust Boundary to Persistent Media




**Srini Narayanamurthy**

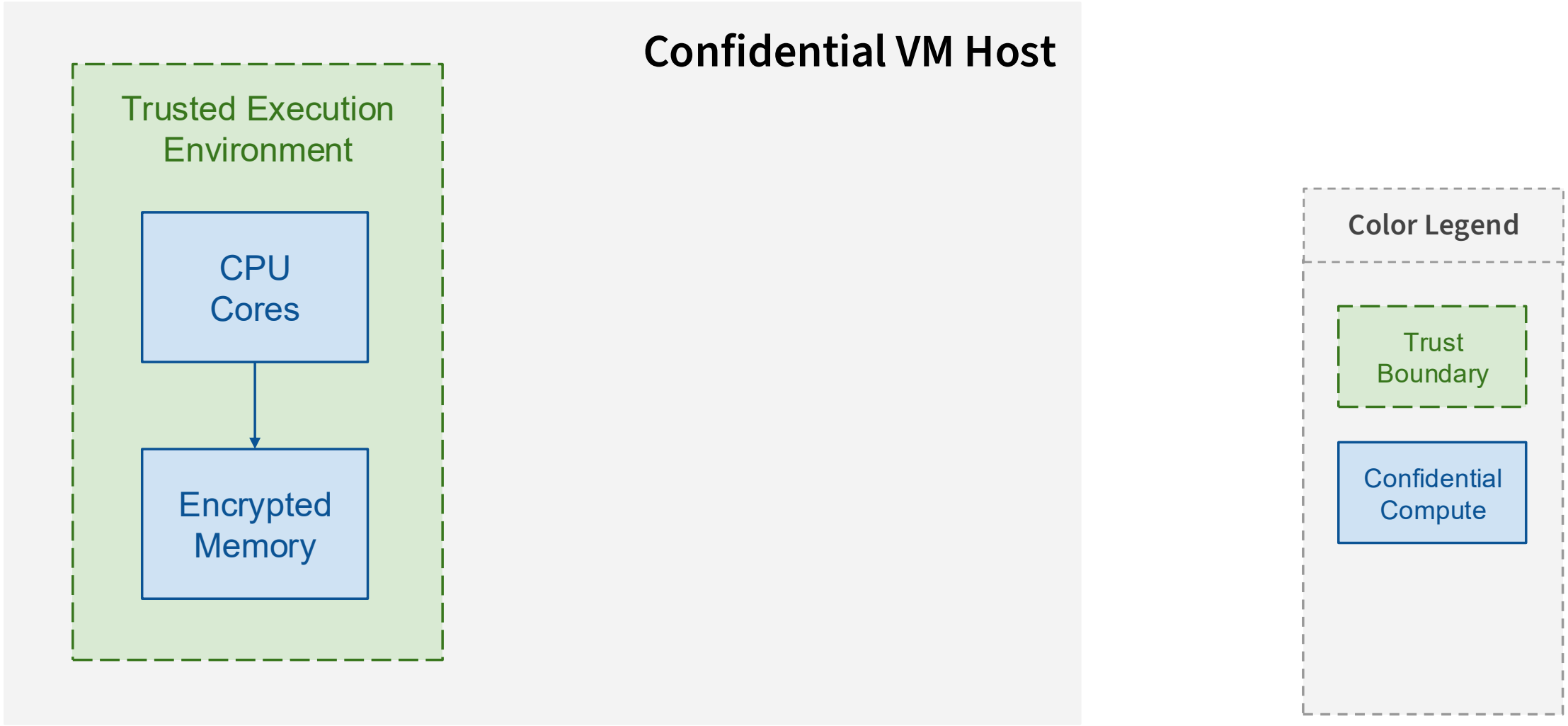
*Staff Engineer, Google*

# Confidential Computing protects data in use

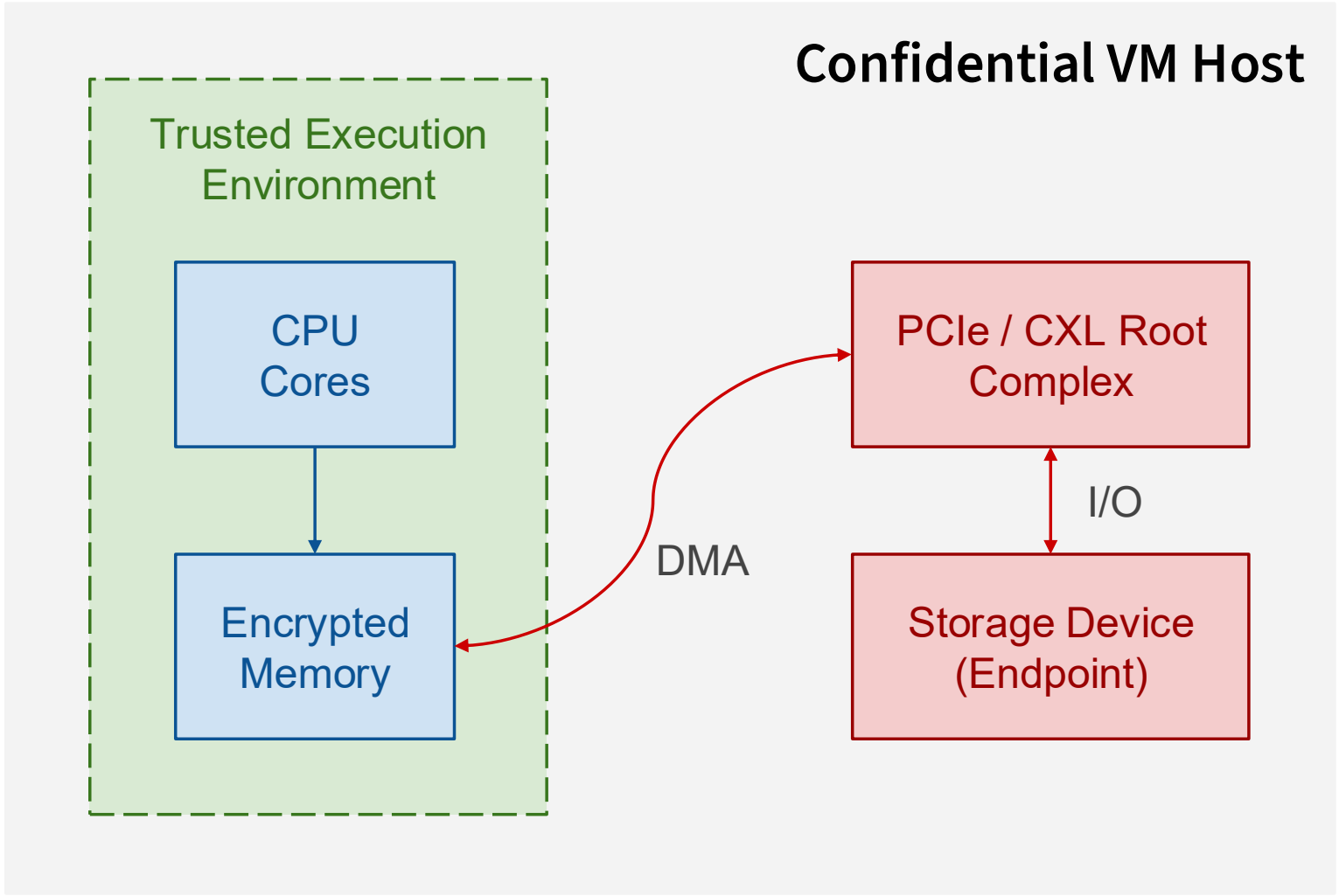
# Confidential Computing protects data in use

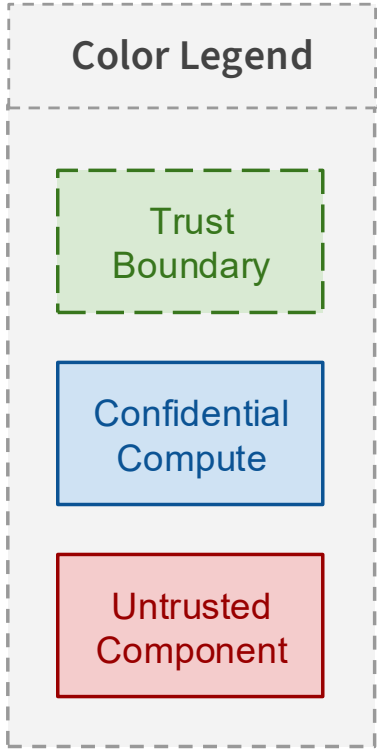
# — then it hits the storage media

# The Open Gate



# The Open Gate



Opening

**Threats** · Evolution

Isolation · Confidentiality · Integrity

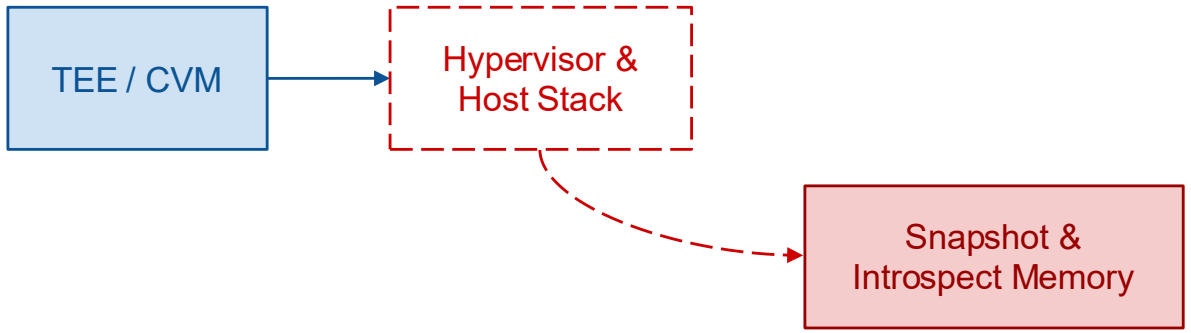
Putting it all together

Alternate Solutions · Reality

Closing

# Every Hop, an Attacker

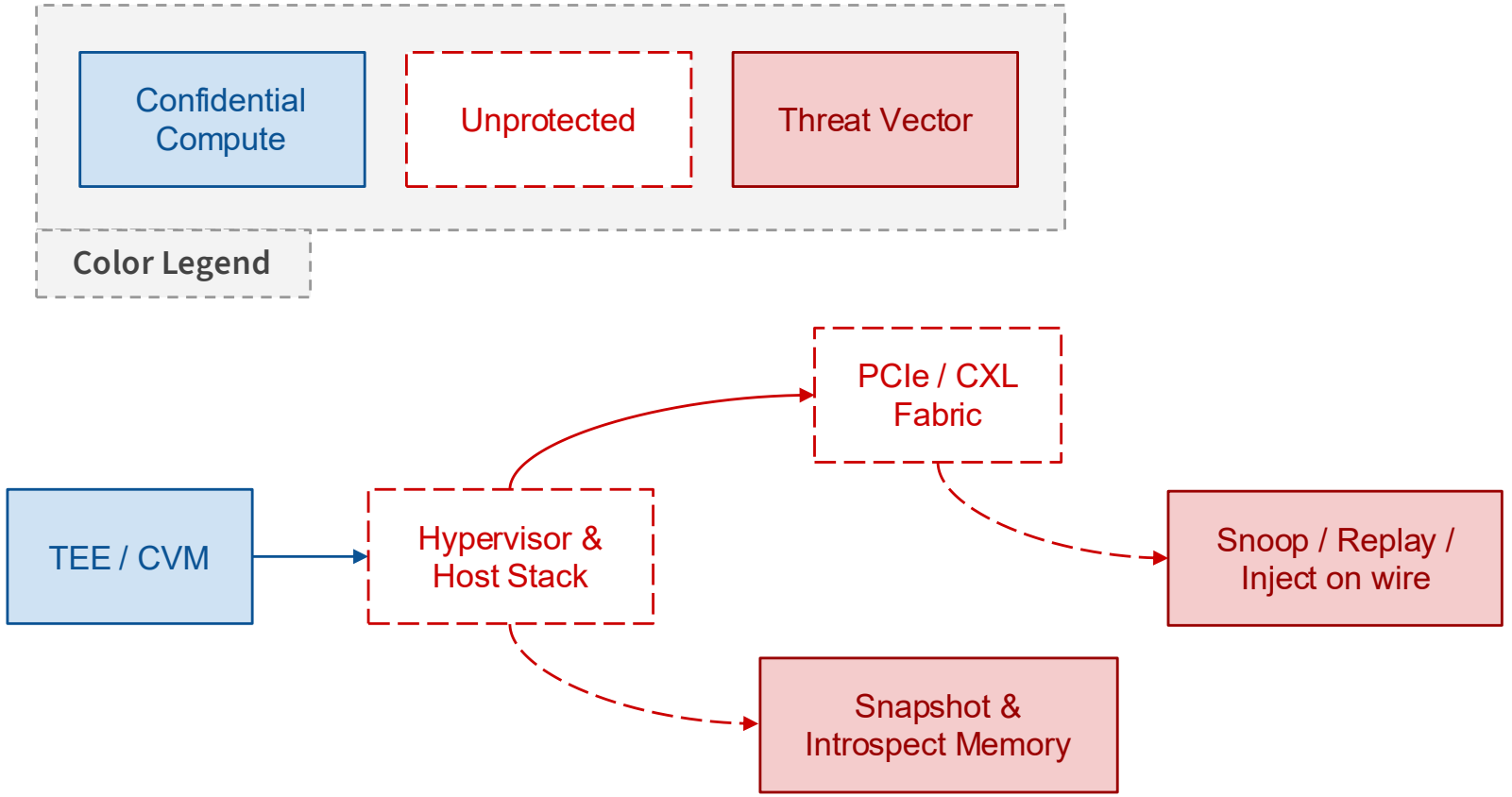



**In scope:** A malicious or compromised operator, hypervisor, or fabric.

**Out of scope:** Invasive silicon attacks and physical decapsulation.

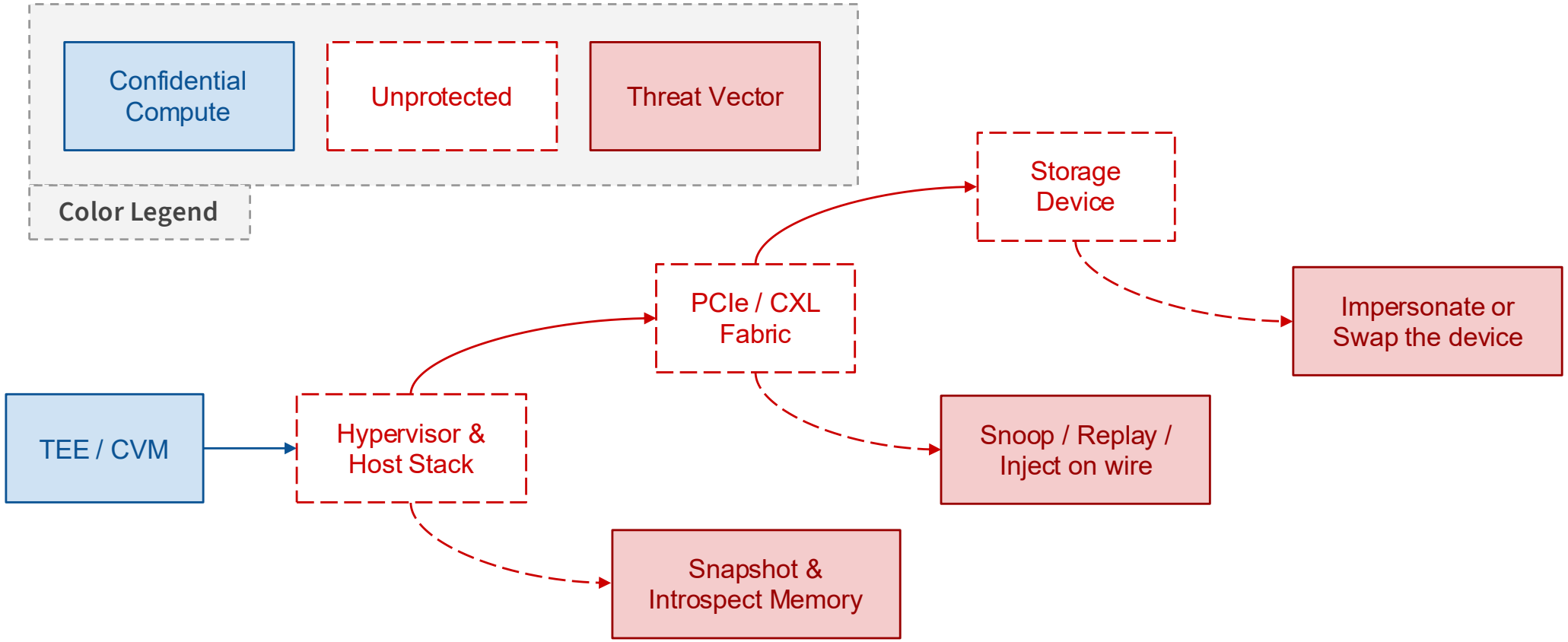
# Every Hop, an Attacker




**In scope:** A malicious or compromised operator, hypervisor, or fabric.

**Out of scope:** Invasive silicon attacks and physical decapsulation.

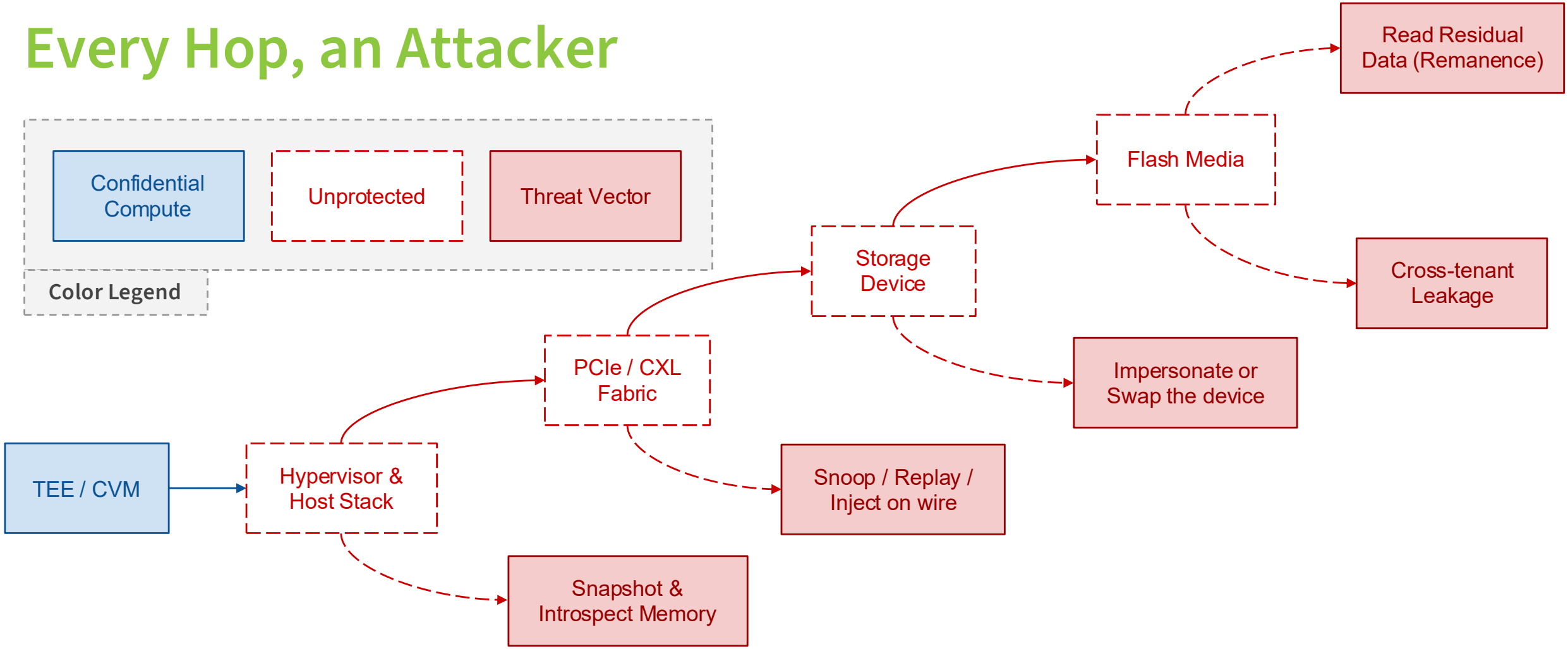
# Every Hop, an Attacker




**In scope:** A malicious or compromised operator, hypervisor, or fabric.

**Out of scope:** Invasive silicon attacks and physical decapsulation.

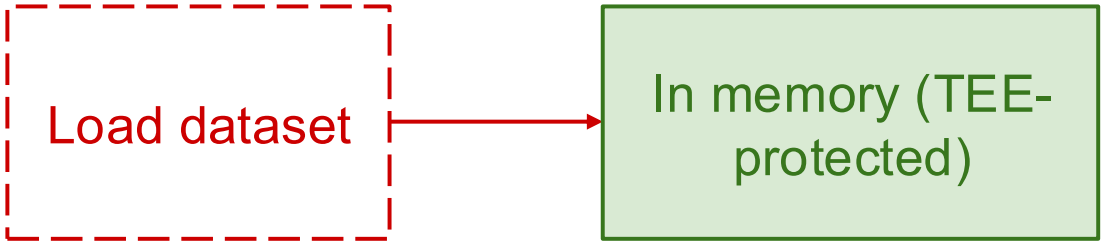
# Every Hop, an Attacker

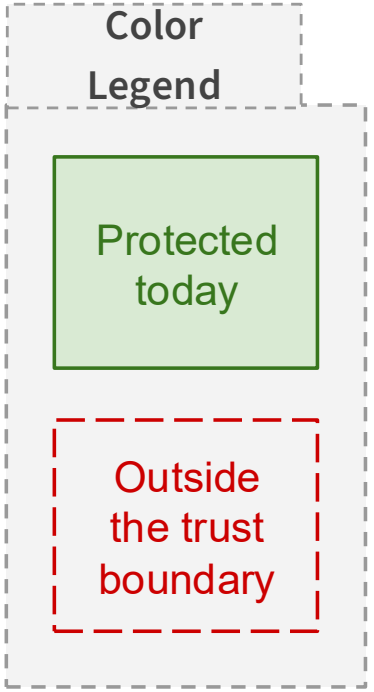



**In scope:** A malicious or compromised operator, hypervisor, or fabric.

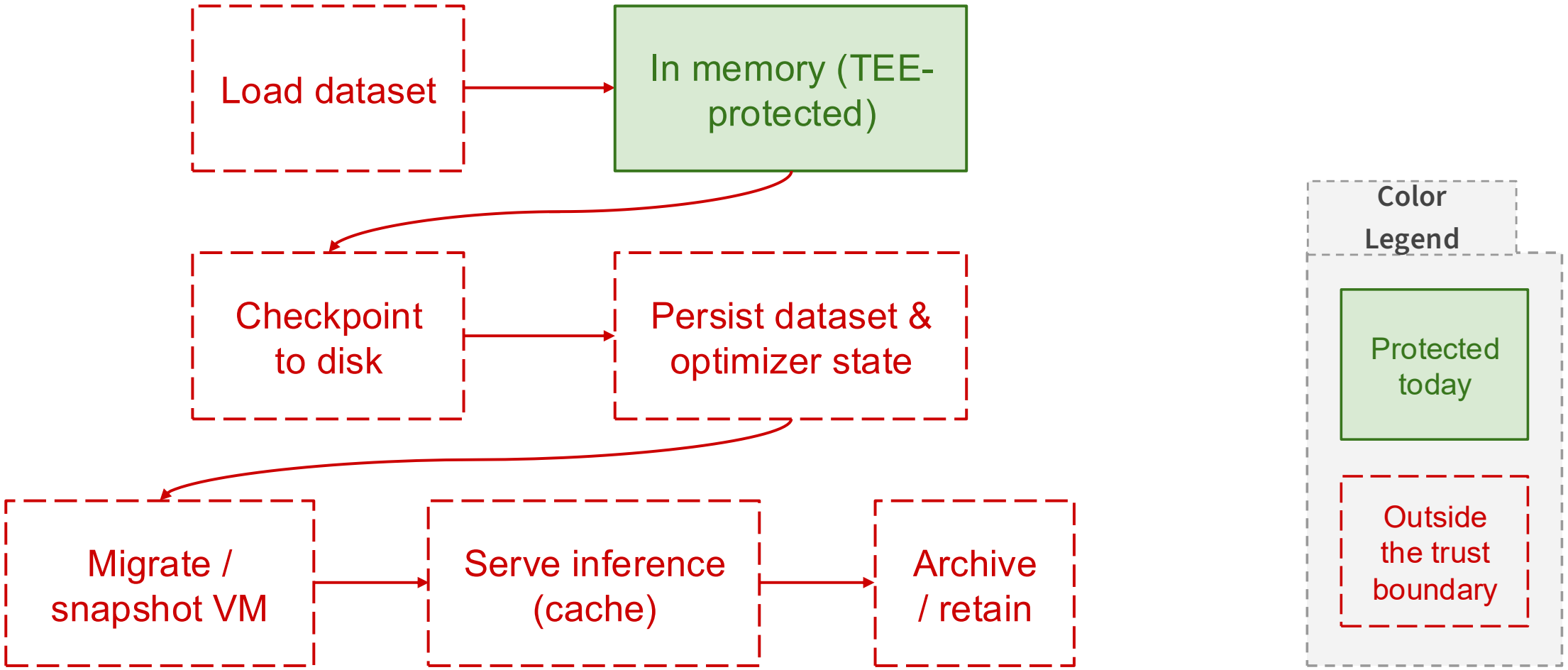
**Out of scope:** Invasive silicon attacks and physical decapsulation.

# One Box Is Safe



# One Box Is Safe



Opening

Threats · **Evolution**

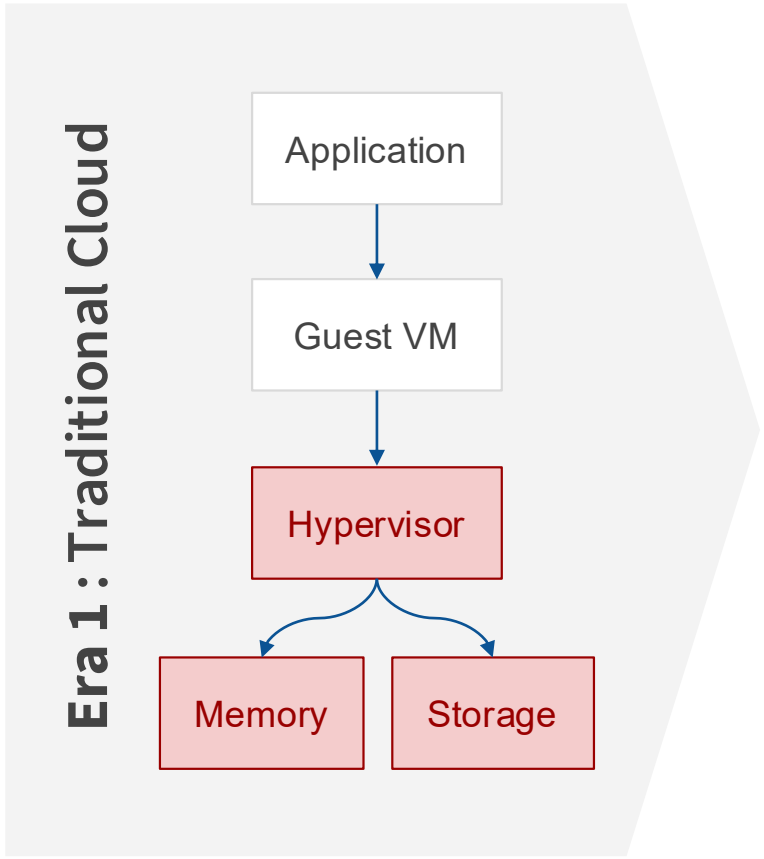
Isolation · Confidentiality · Integrity

Putting it all together

Alternate Solutions · Reality

Closing

# A Boundary on the Move

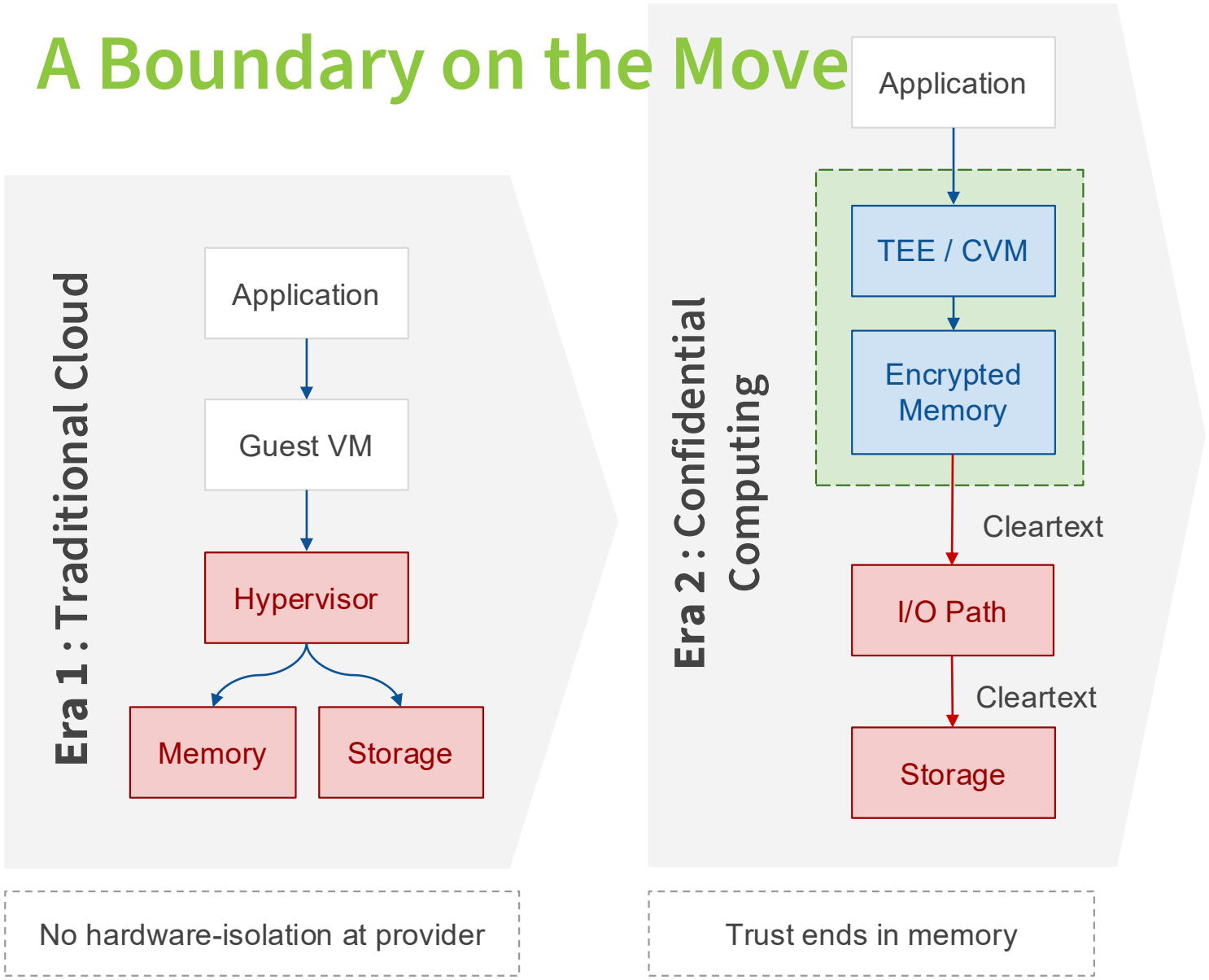



No hardware-isolation at provider

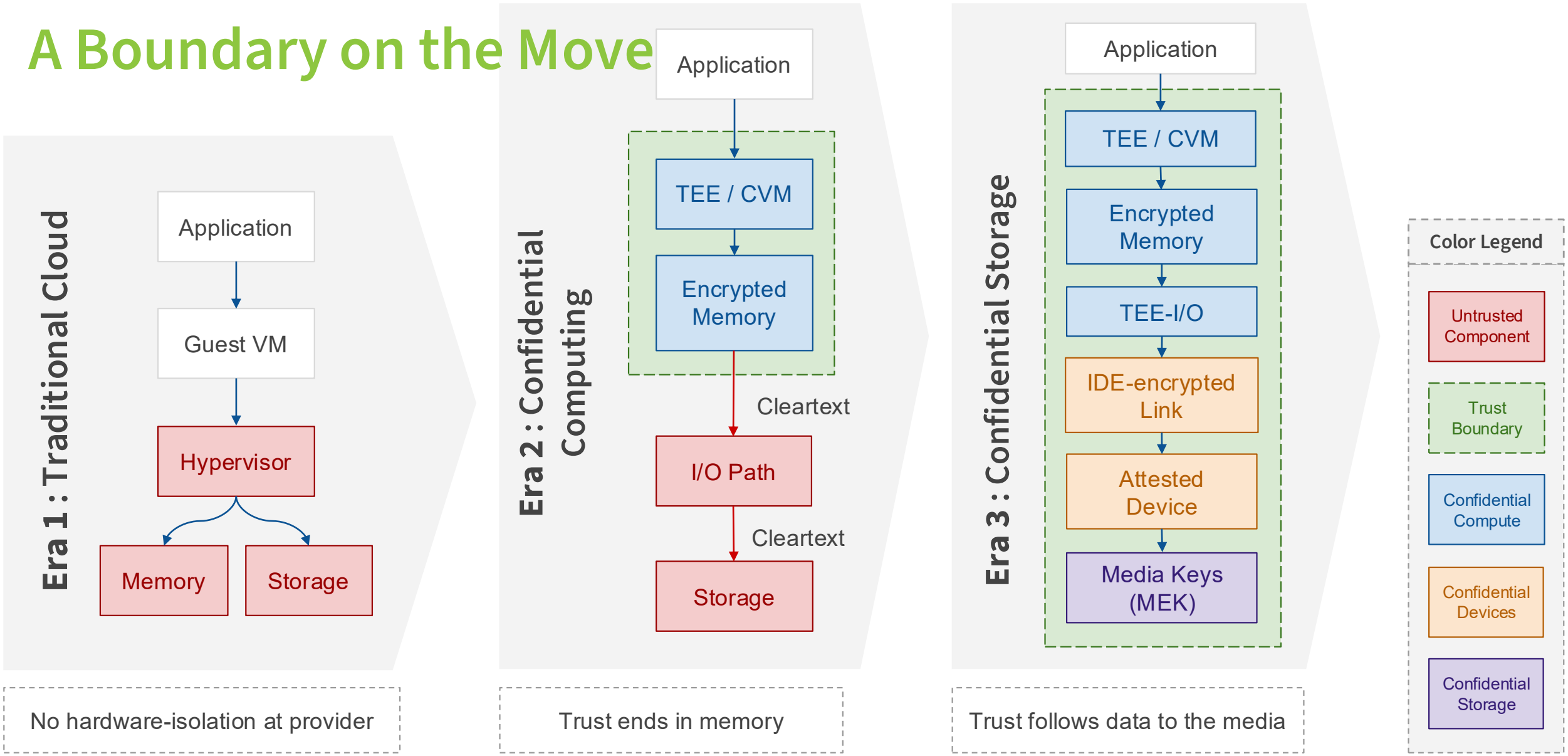
Color Legend

Untrusted Component

# A Boundary on the Move



# A Boundary on the Move



# Three Architectures, One Idea

| Vendor | Confidential VM | Trusted I/O |
|---|---|---|
| Intel | TDX (Trust Domains) | TDX Connect |
| AMD | SEV-SNP | SEV-TIO |
| Arm | CCA (Realms) | CCA Device Assignment |

# Three Architectures, One Idea

| Vendor | Confidential VM | Trusted I/O |
|---|---|---|
| Intel | TDX (Trust Domains) | TDX Connect |
| AMD | SEV-SNP | SEV-TIO |
| Arm | CCA (Realms) | CCA Device Assignment |

Opening

Threats · Evolution

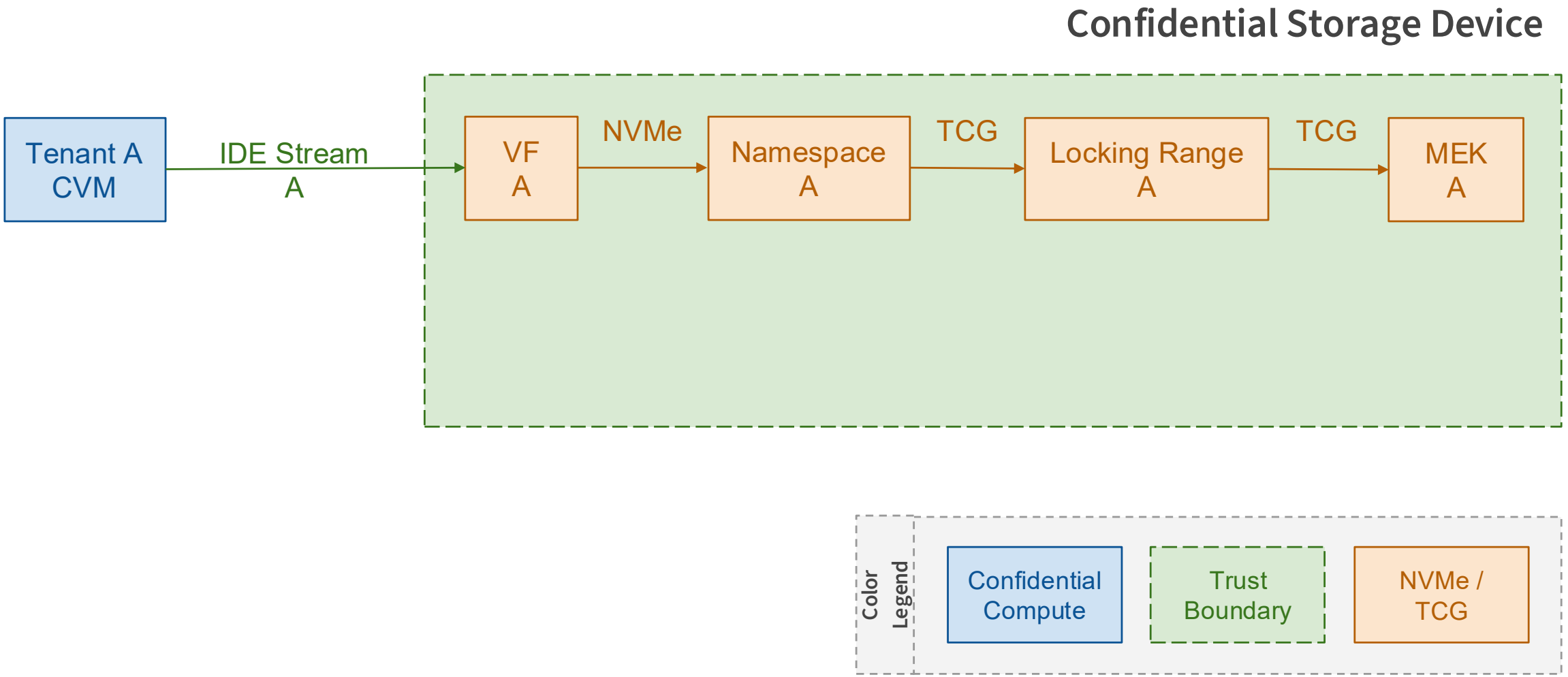
**Isolation** · Confidentiality · Integrity

Putting it all together

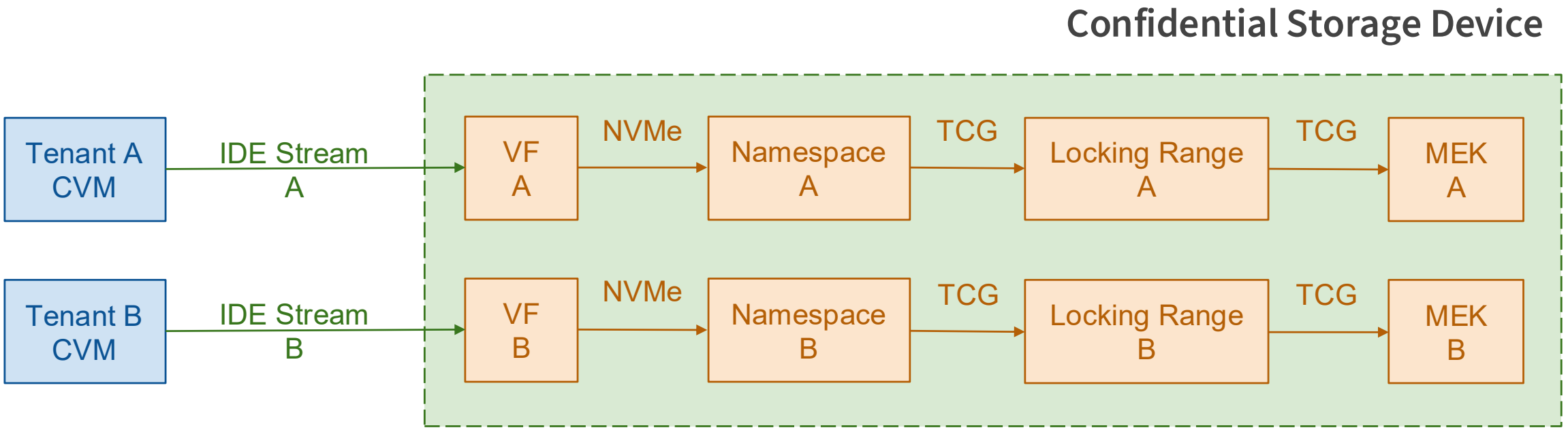
Alternate Solutions · Reality

Closing

# Isolation, All the Way Down



# Isolation, All the Way Down




A separate IDE stream means
one tenant's ciphertext is not decryptable by another
— even on shared silicon.



Opening

Threats · Evolution

Isolation · **Confidentiality** · Integrity

Putting it all together

Alternate Solutions · Reality

Closing

# Keys Follow Ownership

Customer
*(Data owner)*

Cloud Provider
*(Operations)*

Device OEM
*(Endorsement)*

← Parties to a Key

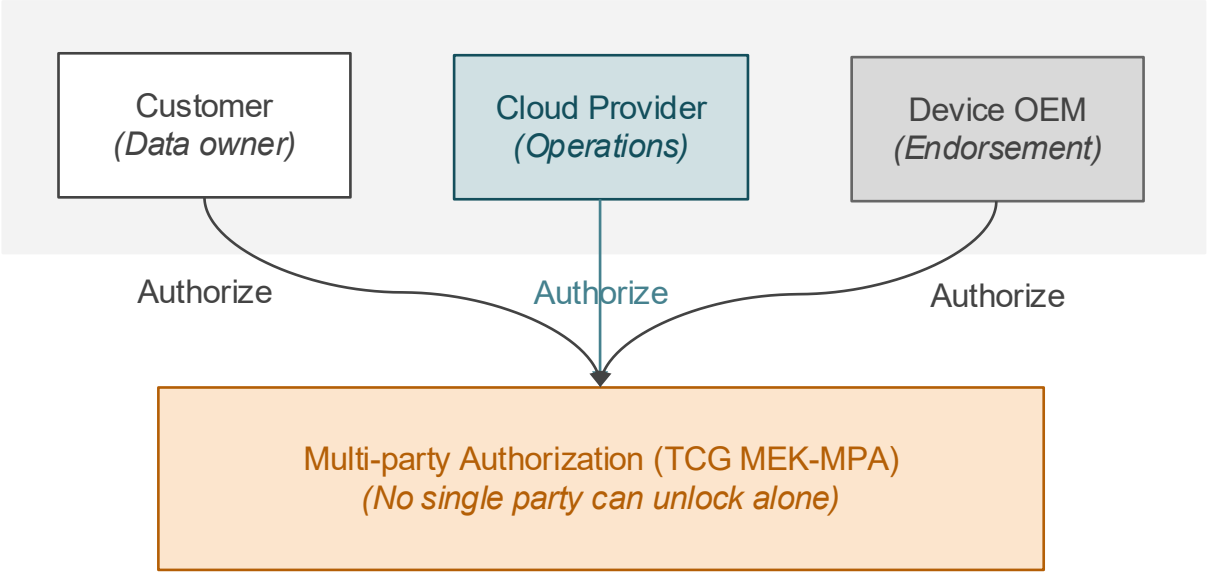
**Color Legend**

Owner's Data / Key

Cloud Key Manager

Device Identifier

# Keys Follow Ownership

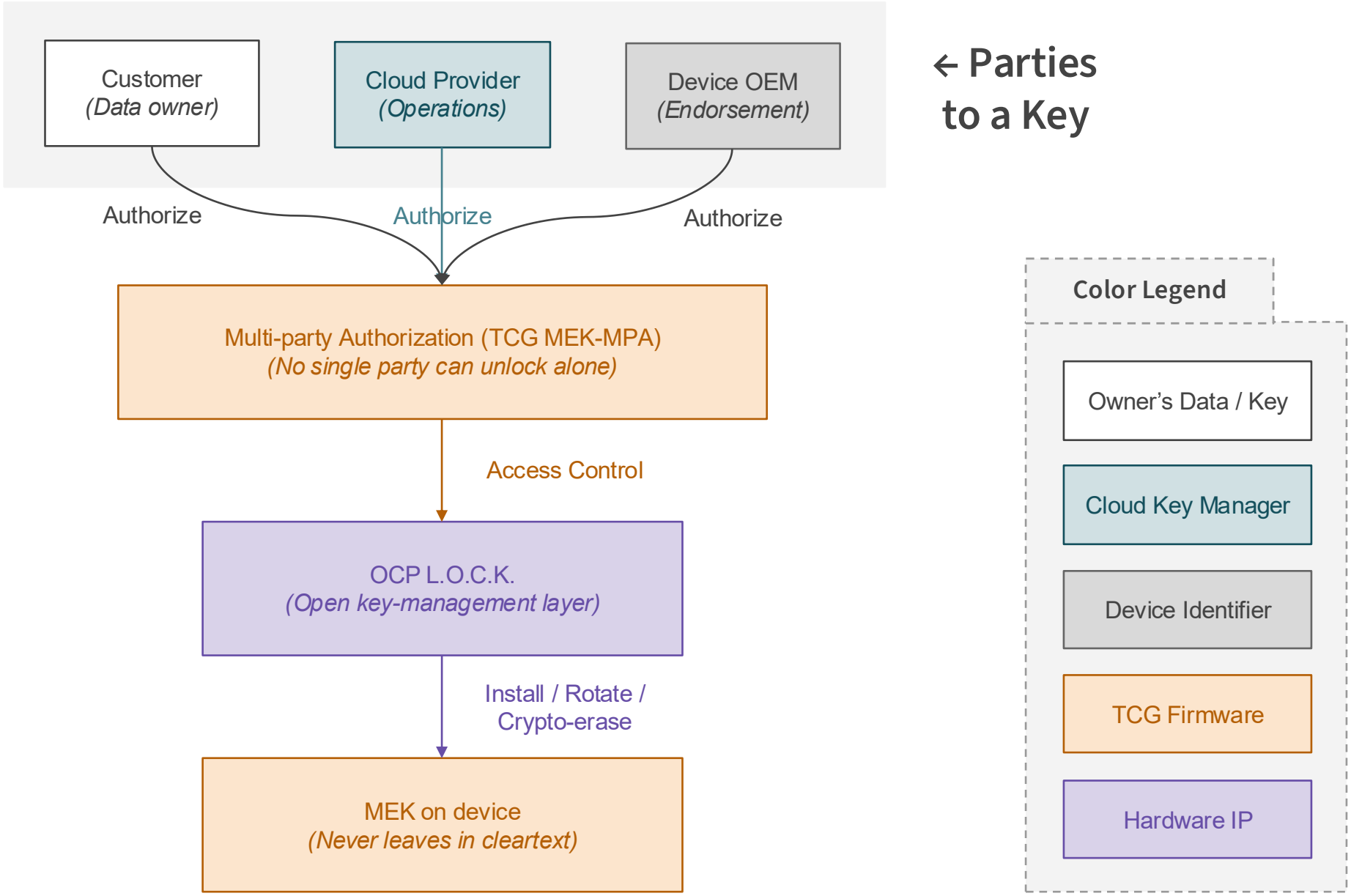



← Parties to a Key

**Color Legend**

- Owner's Data / Key
- Cloud Key Manager
- Device Identifier
- TCG Firmware

# Keys Follow Ownership



Opening

Threats · Evolution

Isolation · Confidentiality · **Integrity**

Putting it all together

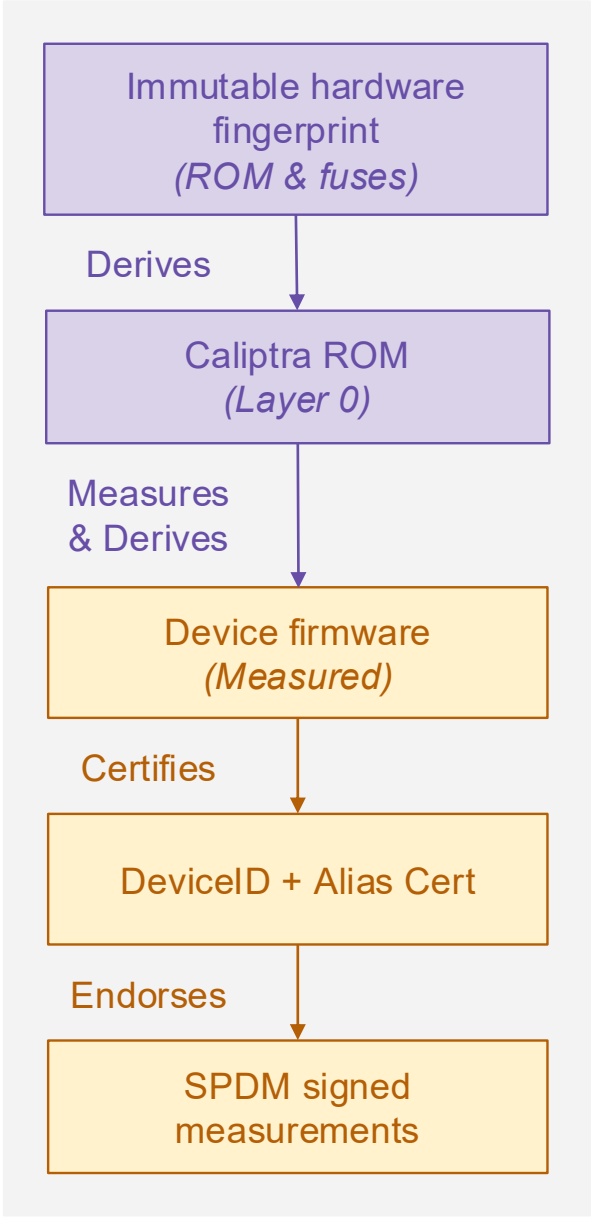
Alternate Solutions · Reality

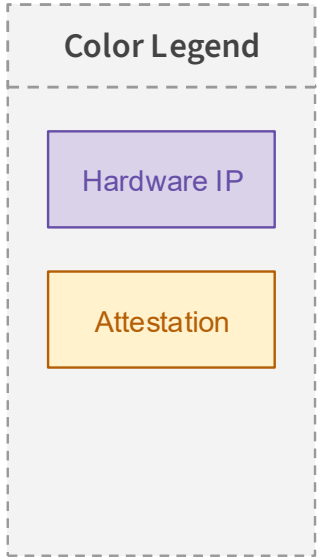
Closing

# Trust, Measured

**Storage Device's Layered Attestation Chain (DICE) →**

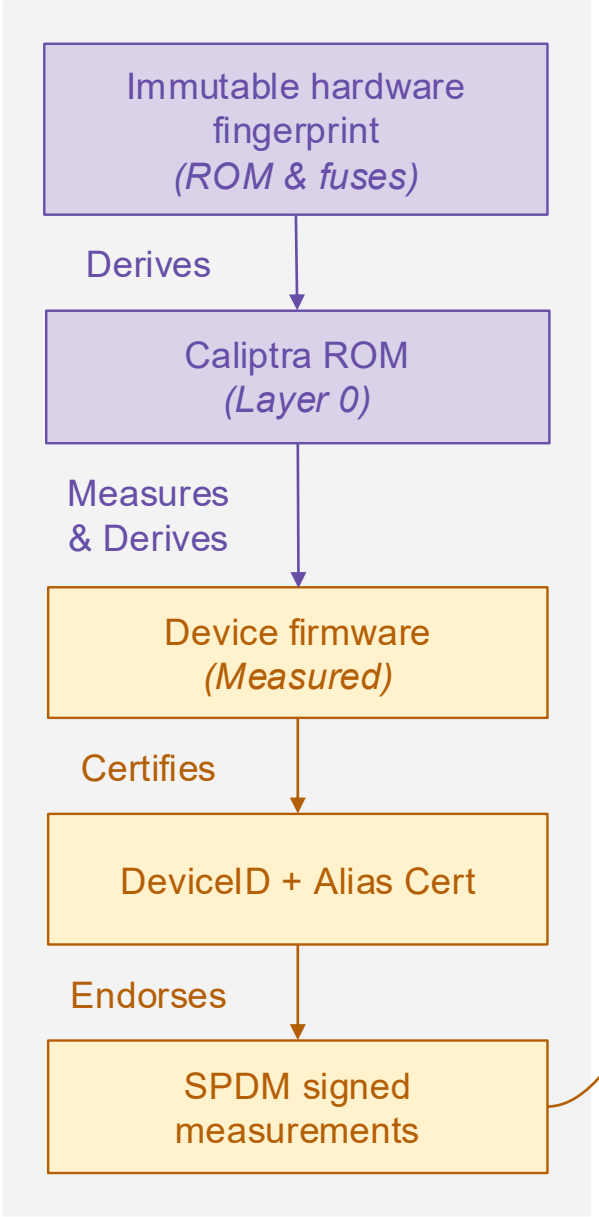
**Each layer measures the next before handing off control.**



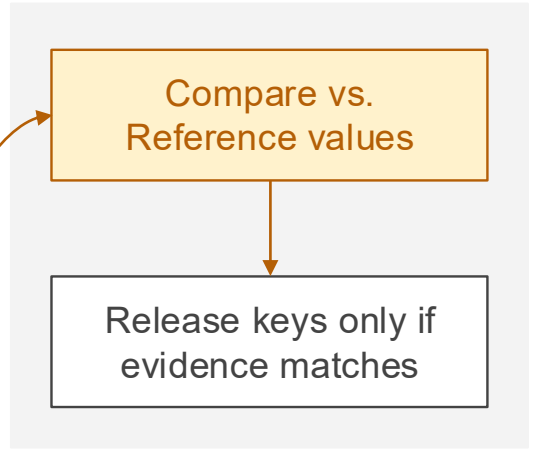



# Trust, Measured

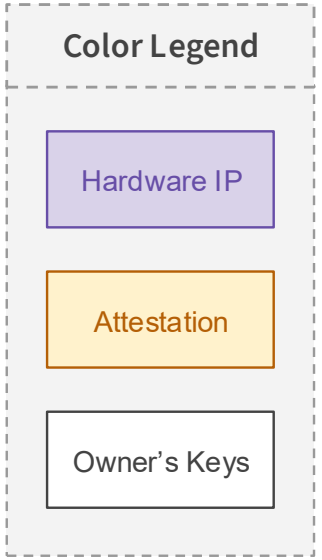
**Storage Device's Layered Attestation Chain (DICE) →**




Evidence




**← Verifier / Tenant**



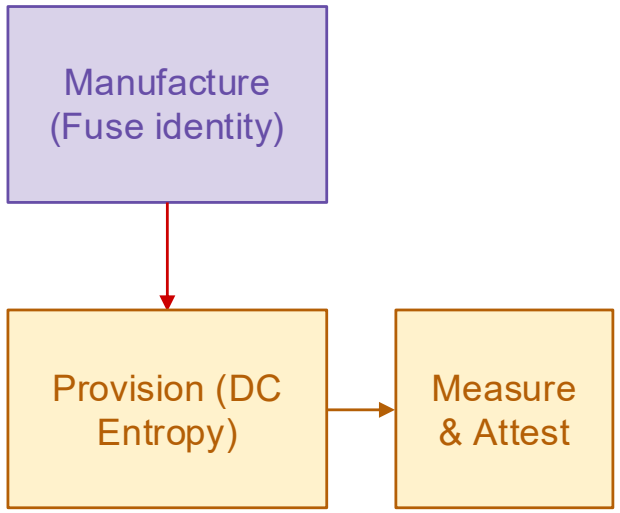

**Each layer measures the next before handing off control.**

**A single signed report vouches for the whole boot chain.**

**Trust becomes measurable.**

# Trust, Continuously Re-Earned

Trust is not a one-time event.
Every device continuously re-earns it

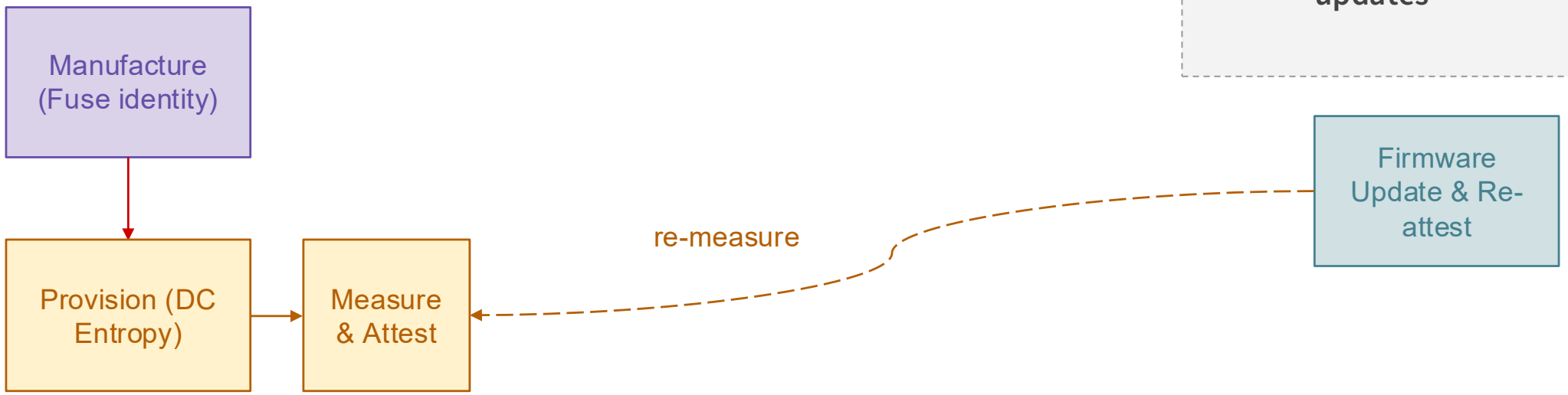



Color Legend

Hardware IP | Attestation | TDISP | NVMe/TCG | Device Lifecycle

# Trust, Continuously Re-Earned

**Trust is not a one-time event.**
**Every device continuously re-earns it across updates**

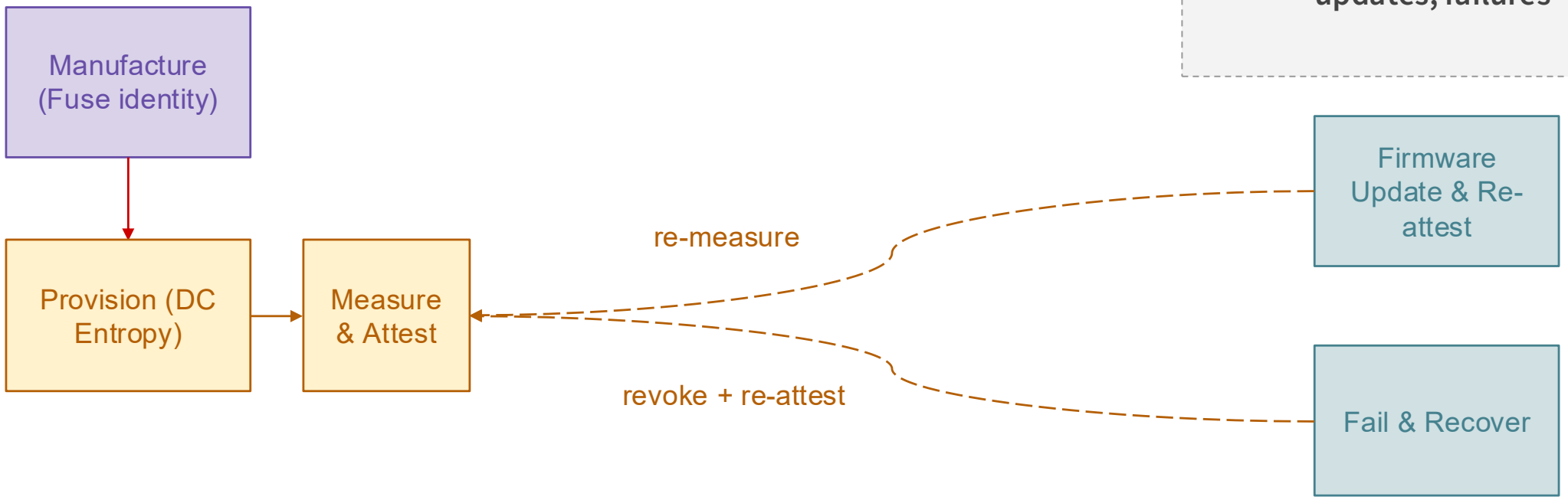



**Color Legend**

Hardware IP | Attestation | TDISP | NVMe/TCG | Device Lifecycle

# Trust, Continuously Re-Earned

**Trust is not a one-time event.**
**Every device continuously re-earns it across updates, failures**

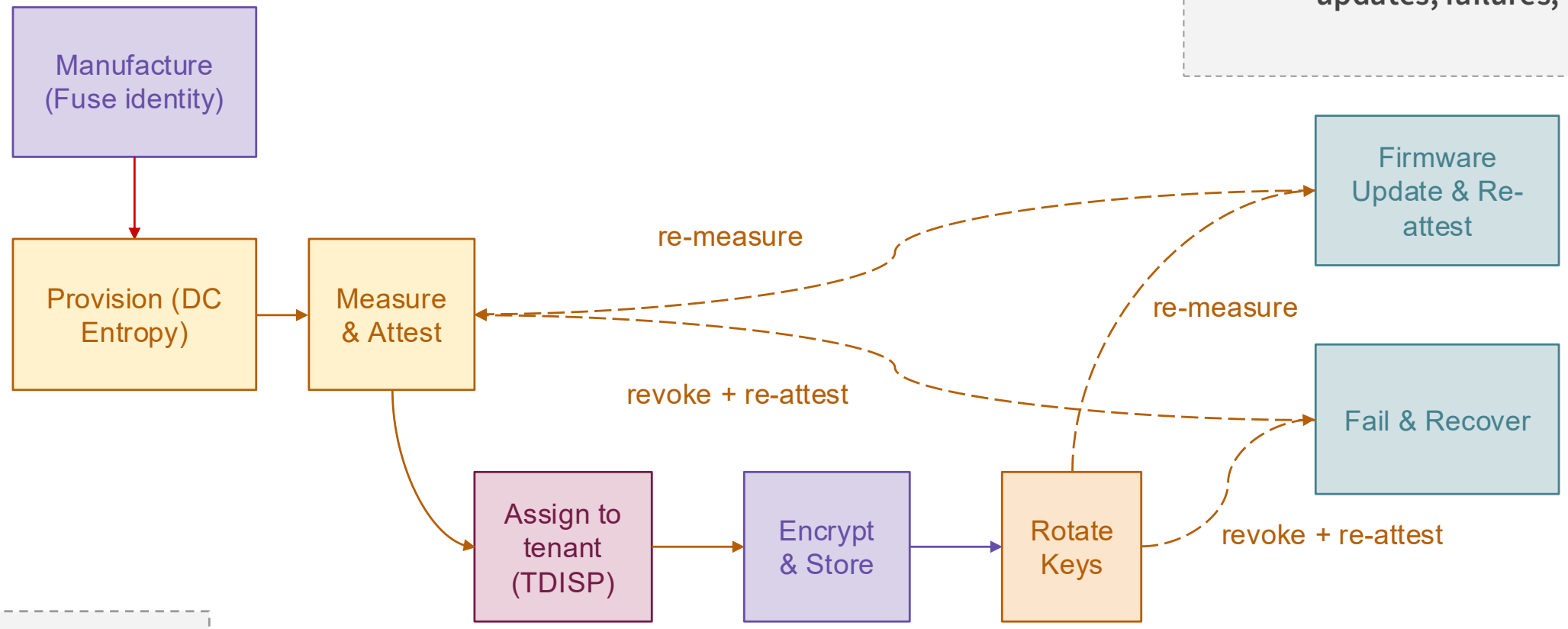



**Color Legend**

Hardware IP | Attestation | TDISP | NVMe/TCG | Device Lifecycle

# Trust, Continuously Re-Earned

**Trust is not a one-time event.**
**Every device continuously re-earns it across updates, failures, reassignment**

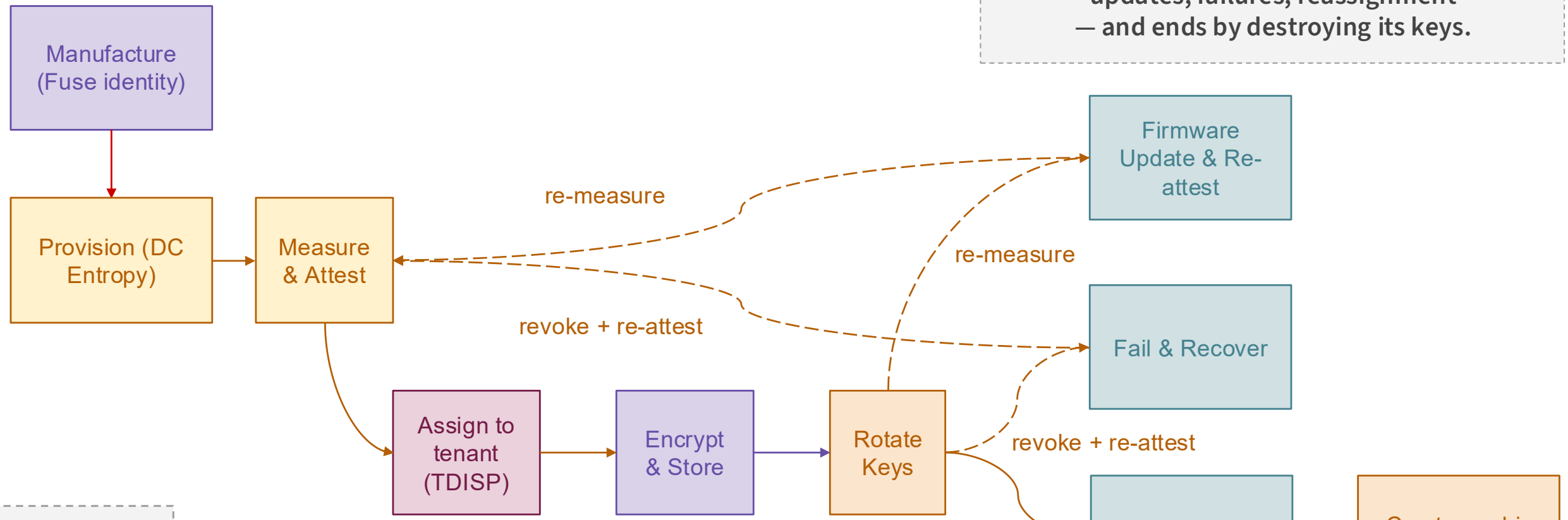



**Color Legend**

Hardware IP | Attestation | TDISP | NVMe/TCG | Device Lifecycle

# Trust, Continuously Re-Earned

**Trust is not a one-time event.**
**Every device continuously re-earns it across updates, failures, reassignment — and ends by destroying its keys.**



Opening

Threats · Evolution

Isolation · Confidentiality · Integrity

**Putting it all together**

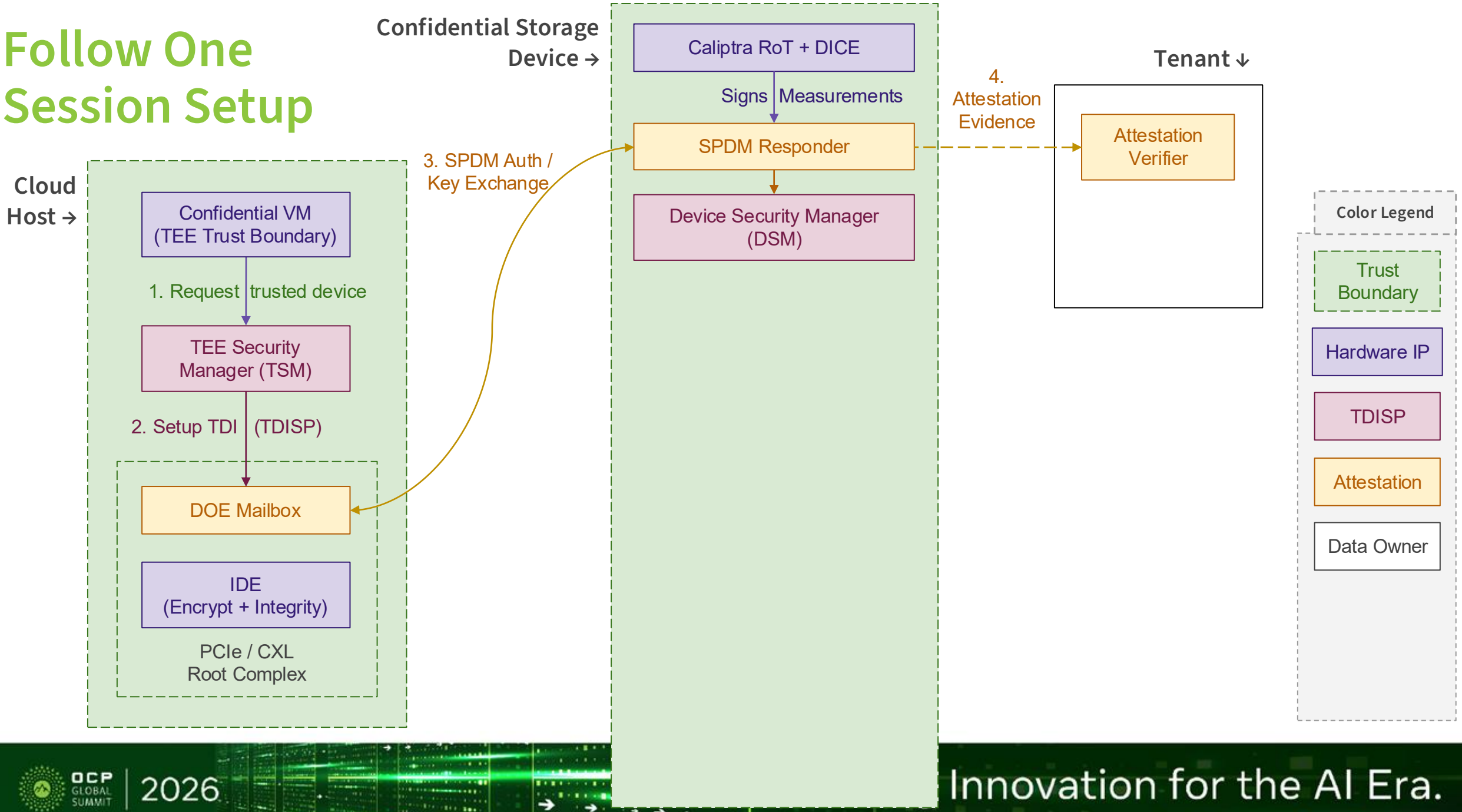
Alternate Solutions · Reality

Closing

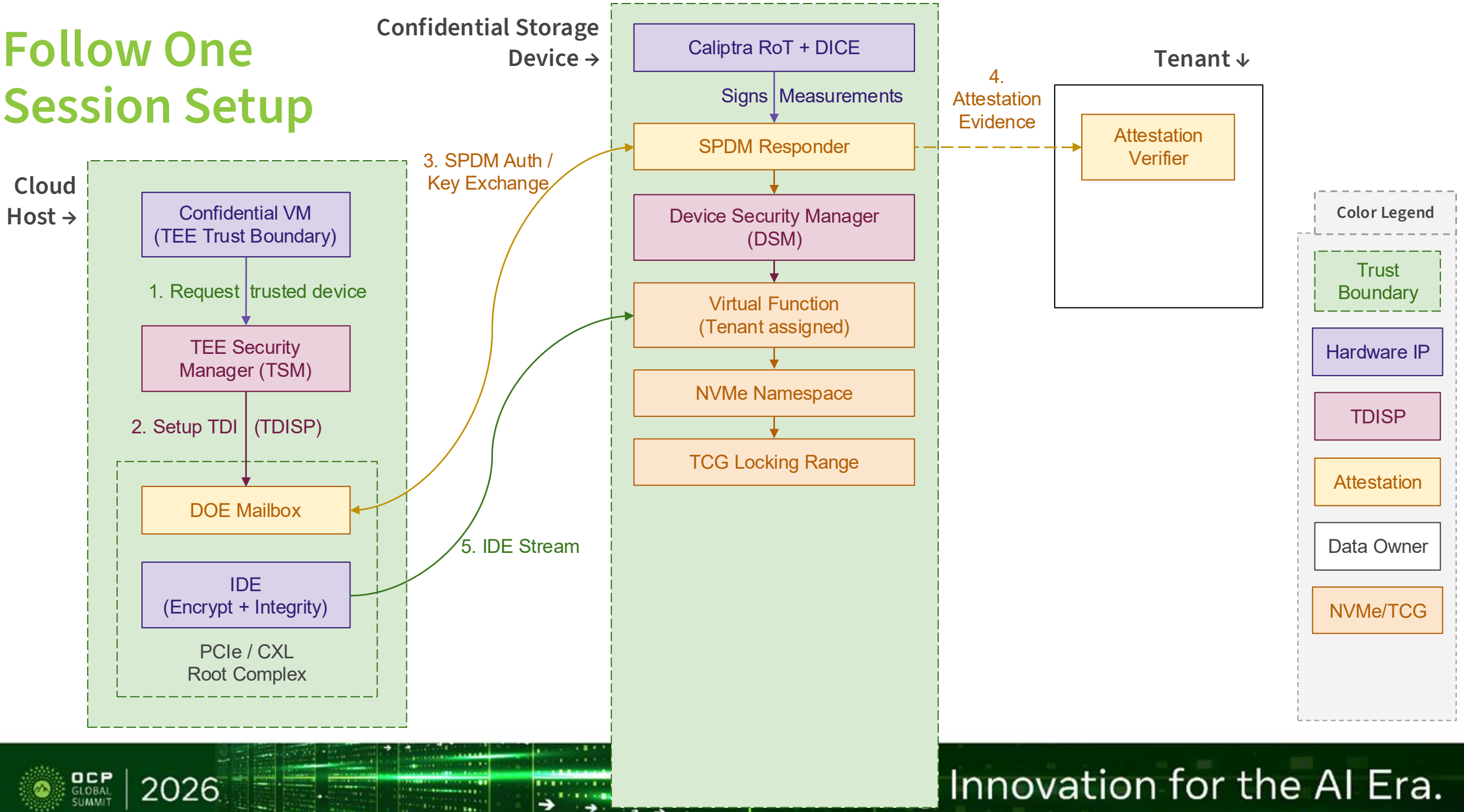
# Follow One Session Setup



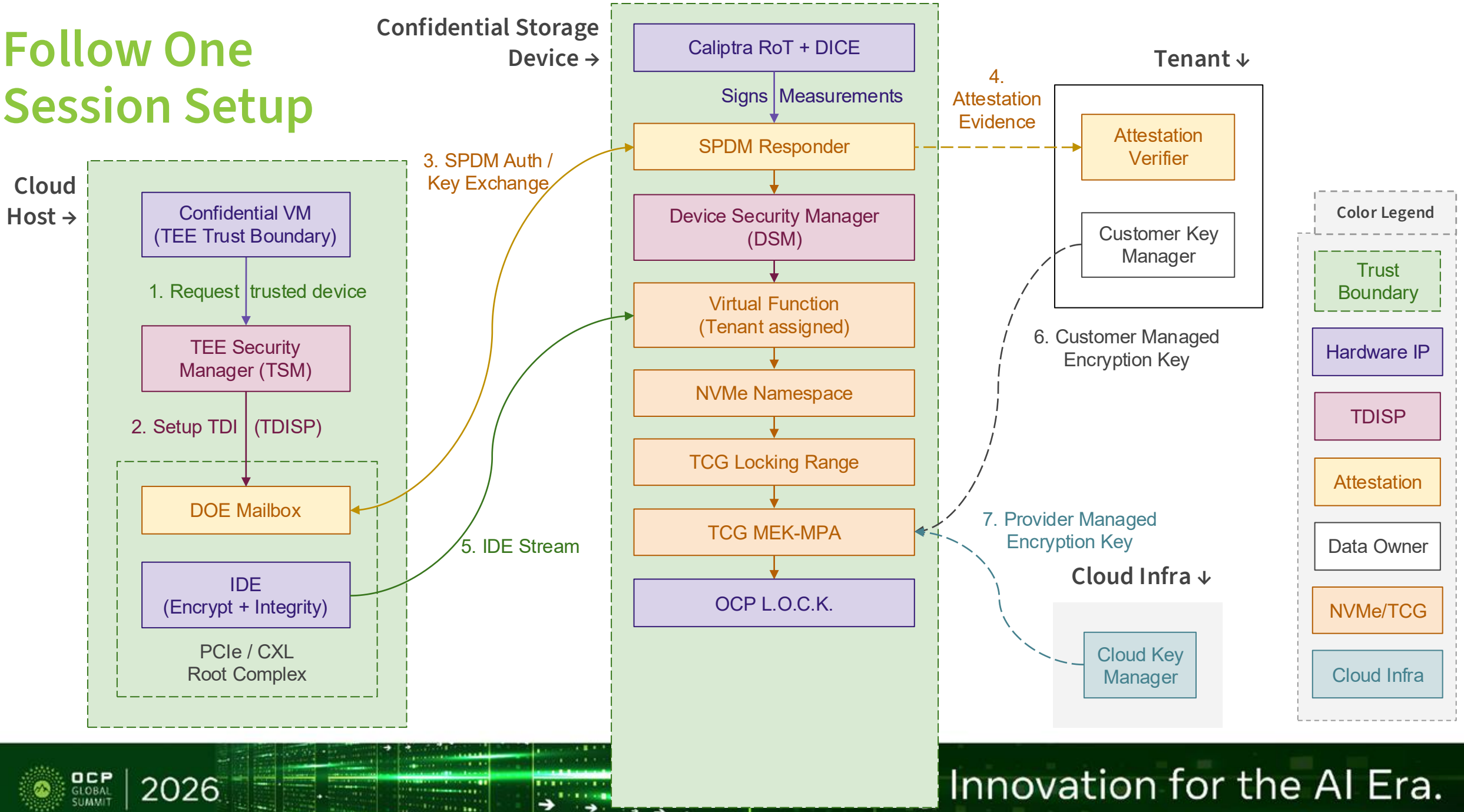
# Follow One Session Setup



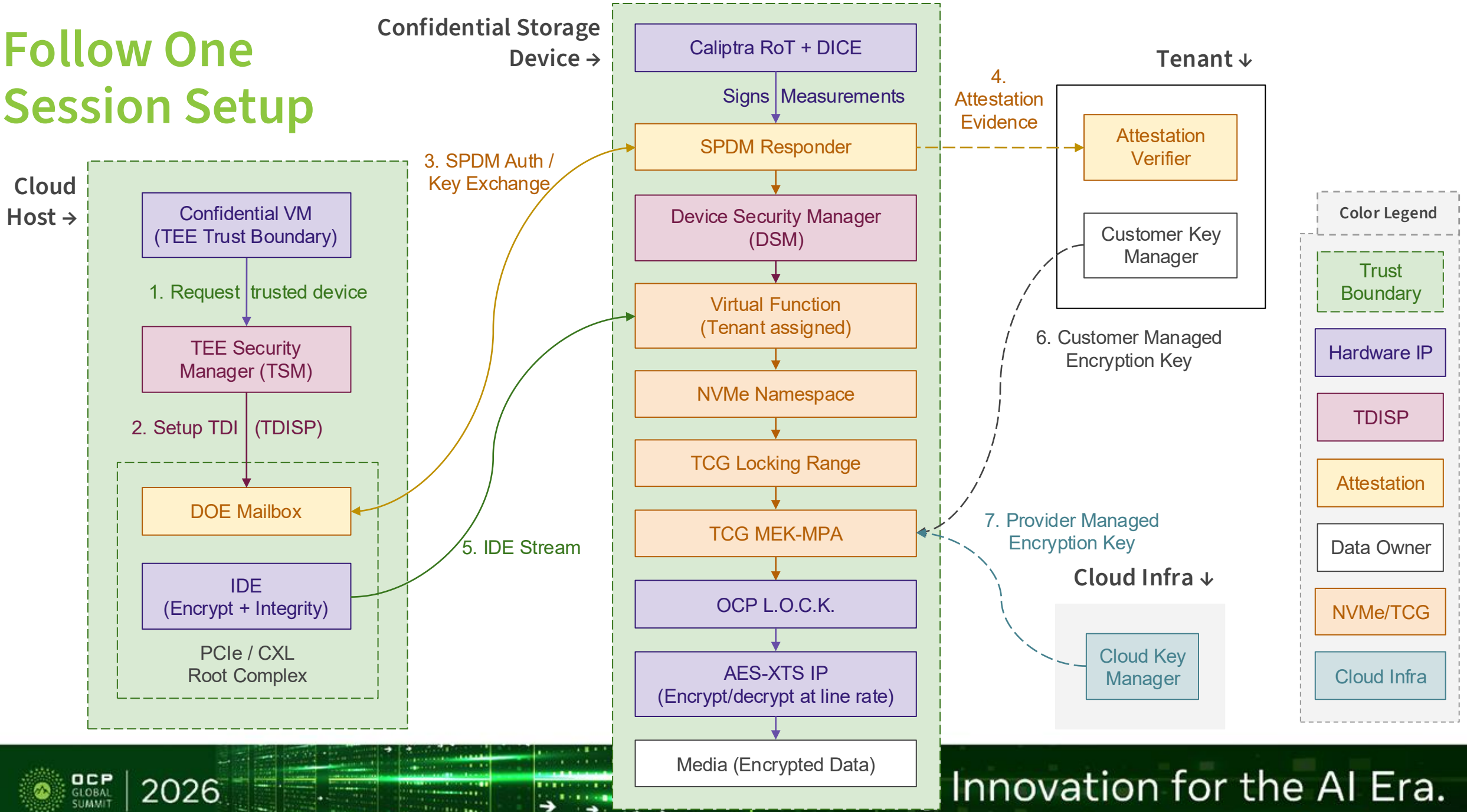
# Follow One Session Setup



# Follow One Session Setup



# Follow One Session Setup



Opening

Threats · Evolution

Isolation · Confidentiality · Integrity

Putting it all together

**Alternate Solutions** · Reality

Closing

# Encrypted Disks Aren't Enough

| Property | Self-encrypting Drive | Host-based Encryption | CSP-managed Keys | Confidential Storage |
|---|---|---|---|---|
| Protects against a **stolen** drive | Yes | Yes | Yes | **Yes** |
| Protects against the **cloud** operator | No | No | No | **Yes** |
| Plaintext hidden from the **hypervisor** | No | No | No | **Yes** |
| Data encrypted on the **wire** (fabric) | No | Partial | No | **Yes** |
| Device identity **attested** to the tenant | No | No | No | **Yes** |
| Data **owner** controls the **keys** | No | Partial | No | **Yes** |
| Cross-tenant **isolation** on shared media | No | No | Partial | **Yes** |

# Why Not Just Encrypt?

| Approach | Overhead / Generality | Root of Trust |
|---|---|---|
| Homomorphic Encryption | Very high; limited operations | Math only |
| Multi-Party Computation | High; bandwidth-bound | Non-collusion |
| Trusted Execution Environment | Near-native; any workload | Hardware + attestation |

# Why Not Just Encrypt?

| Approach | Overhead / Generality | Root of Trust |
|---|---|---|
| Homomorphic Encryption | Very high; limited operations | Math only |
| Multi-Party Computation | High; bandwidth-bound | Non-collusion |
| Trusted Execution Environment | Near-native; any workload | Hardware + attestation |

# Why Not Just Encrypt?

| Approach | Overhead / Generality | Root of Trust |
|---|---|---|
| Homomorphic Encryption | Very high; limited operations | Math only |
| Multi-Party Computation | High; bandwidth-bound | Non-collusion |
| **Trusted Execution Environment** | **Near-native; any workload** | **Hardware + attestation** |

Opening

Threats · Evolution

Isolation · Confidentiality · Integrity

Putting it all together

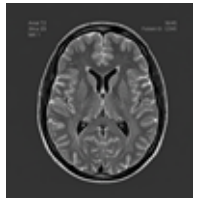
Alternate Solutions · **Reality**

Closing

# AI Makes It Real



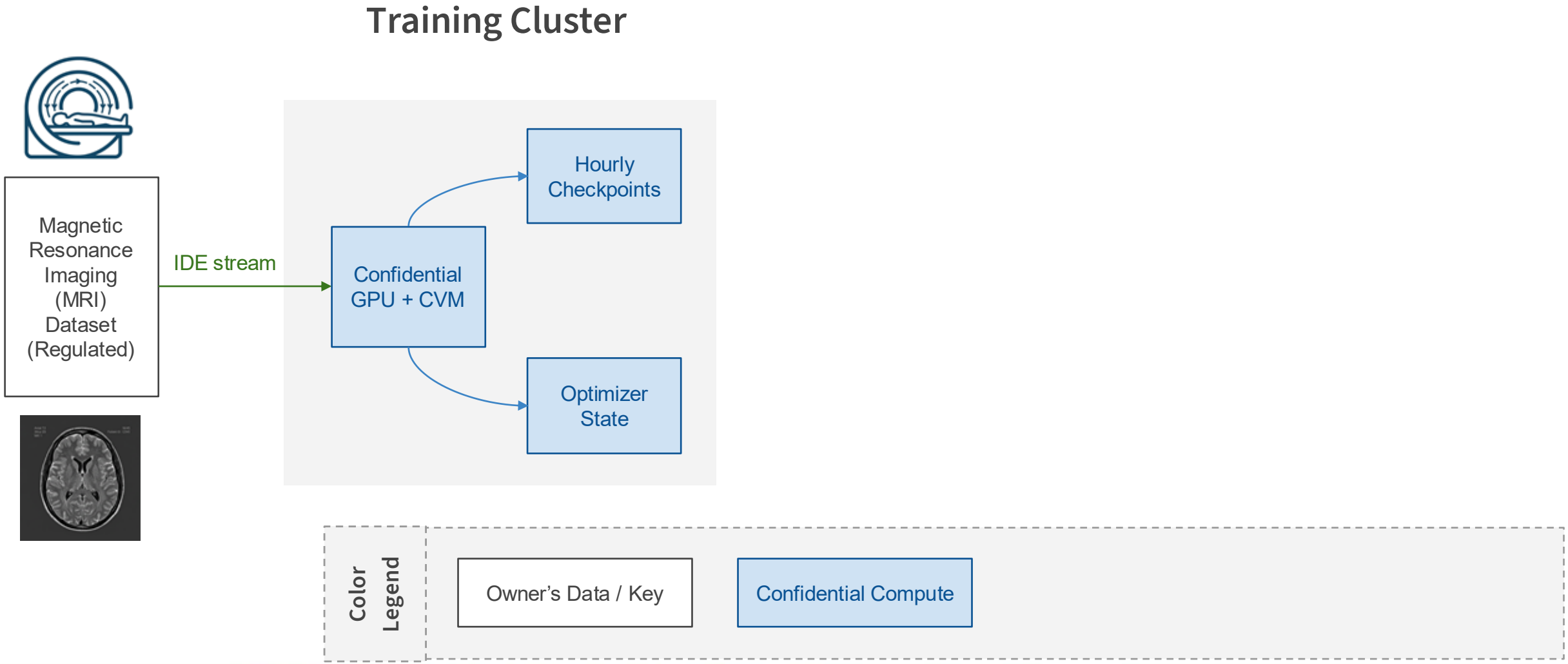
Magnetic Resonance Imaging (MRI) Dataset (Regulated)



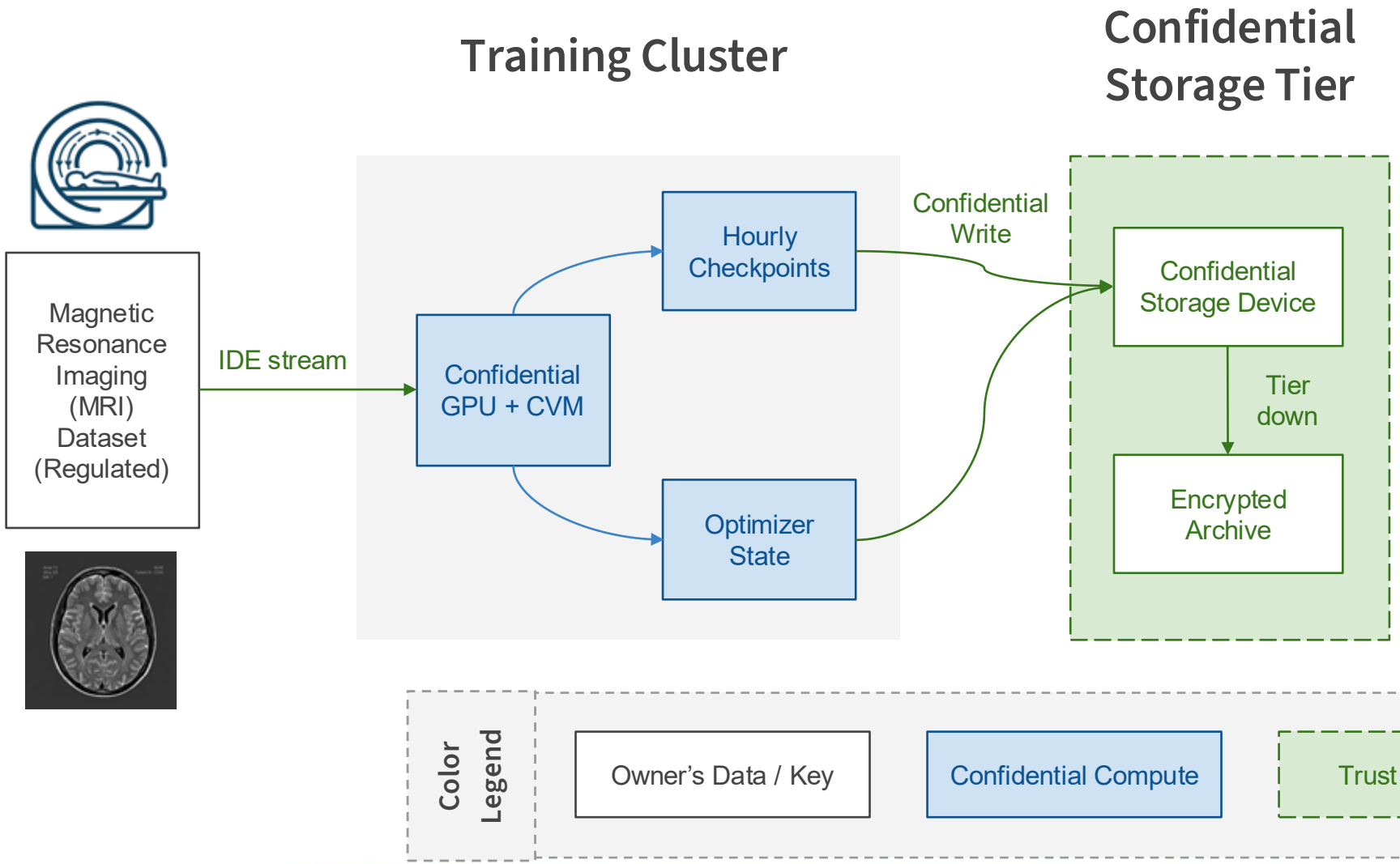
Color Legend

Owner's Data / Key

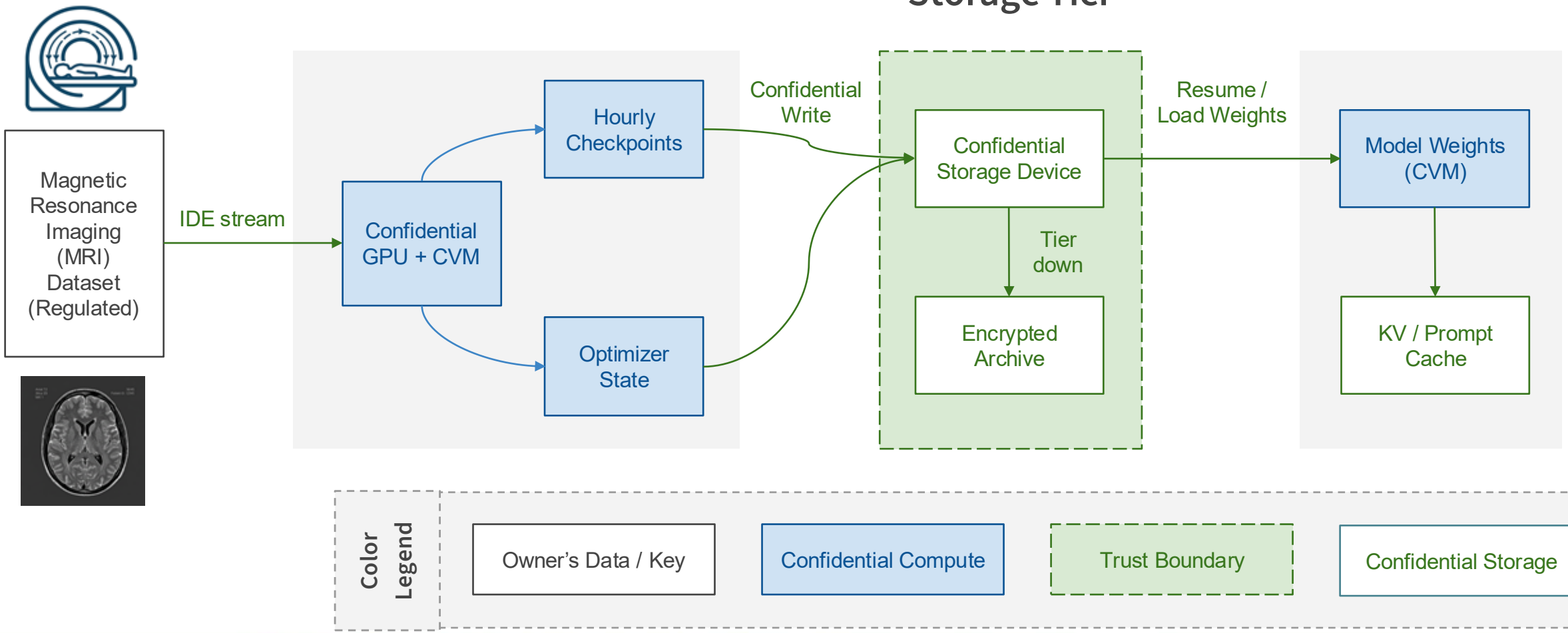
# AI Makes It Real



# AI Makes It Real



# AI Makes It Real



# Reality at Hyperscale

| Challenge | Mechanism |
|---|---|
| **Firmware update** invalidates attestation | Signed update via Caliptra; automatic **re-measure and re-attest** |
| Device **failure** / replacement | Revoke TDI binding; **re-bind and re-attest** on healthy hardware |
| **Key rotation** without downtime | **Quorum** based authorization |
| Tenant offboarding / **decommission** | **Cryptographic erase** of media keys |
| **Attestation** at fleet scale | **Cached reference values**; verifier services; batch appraisal |
| **Link-encryption** overhead | Hardware IDE engines at **line rate**; selective streams |
| Namespace / **key density** per device | Key management; **per-namespace TCG ranges** |

Opening

Threats · Evolution

Isolation · Confidentiality · Integrity

Putting it all together

Alternate Solutions · Reality

**Closing**

# The Last Trusted Party

**Confidential storage removes the last trusted party between the data owner and the media.**

**Confidentiality becomes an end-to-end property.**

**Open frontiers**

**Disaggregation · Storage-side Compute · Confidential Data Services**

# Leverages Existing Standards

| Architectural Responsibility | Standard / Owner |
|---|---|
| CPU / VM Trust | Intel TDX / AMD SEV-TIO / Arm CCA |
| Device Trust | DMTF **SPDM** + **DOE** |
| Fabric Trust | **PCIe** / **CXL IDE** |
| Device Assignment | PCI-SIG **TDISP** |
| Storage Isolation | **NVMe** namespaces |
| Persistent Encryption | TCG **Opal**, **MEK-MPA**, et al. |
| Hardware Identity | **Caliptra** + TCG **DICE** |
| Fleet Key Ops | OCP **L.O.C.K** |

# Thank You!

## Call to Action

### Contribute to Open Standards

**OCP :** Caliptra · L.O.C.K

**TCG :** DICE · Opal · MEK-MPA

**DMTF :** SPDM + DOE · PCIe / CXL IDE · PCI-SIG TDISP

**NVMe**